

maps of India into Google Maps to make them useful out of the box, without a net connection. The same goes for search and audio and video – which should download and store songs and videos that were liked, so that the associated apps are not totally useless when there's no connectivity.

Apps that allow local connectivity and can be of some use will also become popular in rural India. Something as simple as connecting two devices together over Wi-Fi Direct or Bluetooth for a simple multiplayer game would be popular. Even better would be payment apps that can work without the internet and allow people to pay one another, and then update accounts when connectivity was restored. There would be problems that need to be sorted out, but it's certainly possible...

BLOCKCHAIN

Remember that offline payment idea we just mentioned? That would need blockchain at some level to even be possible at all. Blockchain is going to be huge for the next billion, especially when it comes to servicing people in remote villages. It's not all financial stuff either, because blockchain is available everywhere from security to gaming and especially all peer-to-peer (P2P) applications. Absolutely everything we mentioned in the earlier point about offline internet will be made more secure and foolproof by using distributed ledger technologies (DLT = blockchain).

AI

This one is kind of a no-brainer because AI is just going to take over everything, so why not the next billion as well? Stuff like the semantic web can only be made a reality when we have better AI that's capable of understanding language. From what's typed in, to the spoken word, and in as many different languages as humans speak, we need the semantic web to make the experience seamless, and feel "real". Here, real would be akin to talking to a smarter human about something and trying to learn from her or him. Then there's big

data, and sorting through it all to help people. If there's one thing that gains a loyal fan base, it's being of genuine help in times of need or desperation. AI can be that saviour who rescues us from ourselves, and gives us the answers we need, such as when to sow seeds, when to expect rain, reminding you not to use an account to pay someone because a regular payment is coming up, etc. Heck, it sounds awesome for us to be a part of as well, so it's not just the next billion who will benefit.

IIOT

If anything, the Internet of Things will be even more important to the next billion than it will be to most of us.



A small cell antenna in Bangalore, deployed to increase capacity in crowded areas.

Let's face it, at best your interactions with IoT will be because of your car or washing machine, or some other useless gadget in your house. Sure, it's great to have your smart fridge remind you to buy more milk, but it's not even in the same category as a warning device informing a farmer that rodents are eating up his crops!

Helping people track goods, protect their livelihoods, inform them when the well they spend four hours a day walking to, just to get a day's supply of drinking water has dried up, so that they don't

waste their time. The IoT is all about practicality, and that's a much bigger deal in rural India than it is in the better connected cities.

5G

The promise of 5G is that it will be much, much faster, and much less congested than 4G currently is. The problem with 5G is that it has limited coverage per base station, and is expensive to set up. However, we expect the push to upgrade to 5G to trickle down 4G base stations to more rural areas, which will end up speeding up the whole network. Plus there are low band versions of 5G (sub-6GHz) that can be deployed in rural areas as there are less structures to interfere with the signals. This means relatively slower speeds but it will still be faster than LTE, and also have much lower latency. For something like real-time conversations with assistants, latency is much more important than bandwidth. If you've ever used "Ok Google" and asked it to do something, and been a tad annoyed by the pause before it does anything... that's latency at play. Also remember that many IoT devices need low latency connections to function at their best, and IoT pilot projects and services are one of the big reasons the next billion will want to come online for.

FIBRE

Honestly, there's just nothing better than fibre. Data transfer at the speed of light, with bandwidth capabilities rising into the terabit ranges, there's really no better way to connect to the Internet. Fibre also has some of the lowest latencies around, and can be laid over large distances without any loss in signal quality. You could lay a single fibre over 60 km long and there would be no loss in quality of signal or bandwidth, and latency would be minimal as well. The only problem with fibre is that it's vulnerable to physical damage, and can be a pain to replace if not planned properly. In rural India, monitoring a fibre installation to make sure no enthusiastic farmer with a shovel digs it up and destroys